Engineering Curves: Construction of Epicycloid

Rolling Circle Method

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Problem Statement

Given
Radius of fixed circle:

$$R = 50 \text{ mm}$$

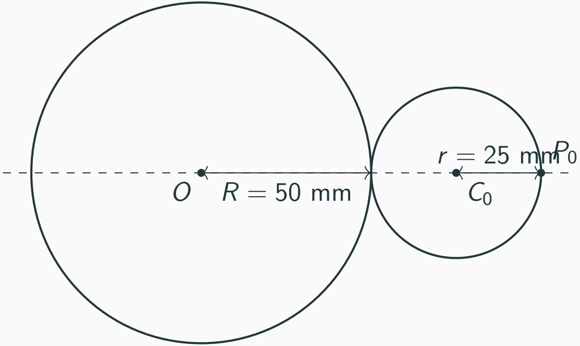
Radius of generating circle:

$$r = 25 \text{ mm}$$

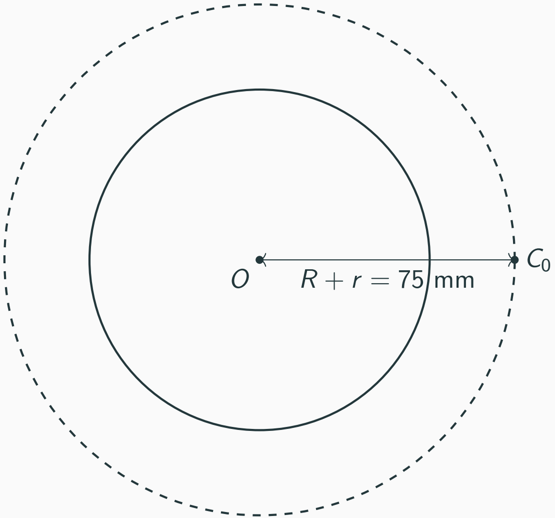
The generating circle rolls externally without slipping around the fixed circle.

Required
Construct the epicycloid for one complete revolution of the generating-circle centre and draw the tangent and normal at a selected point P.

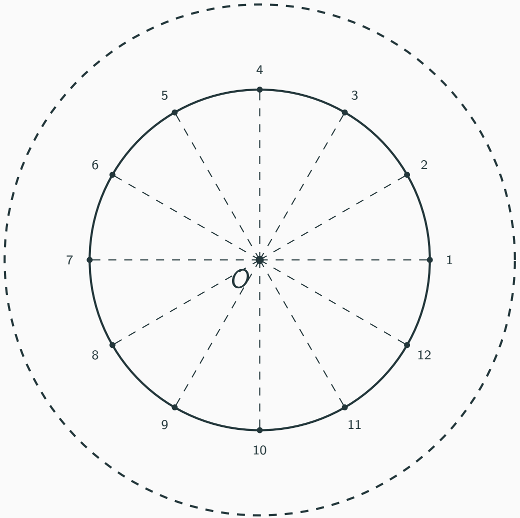
Step 1: Draw the Fixed Circle and Generating Circle



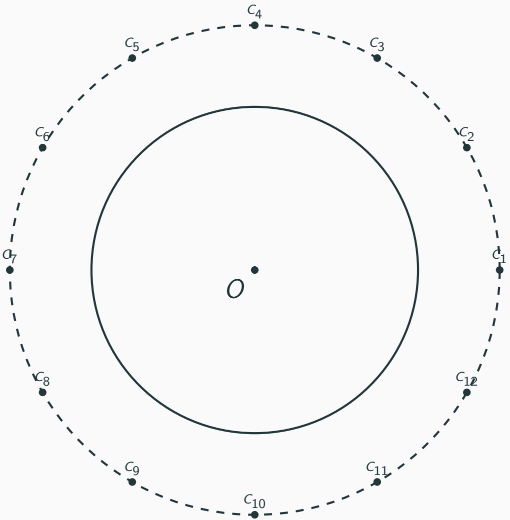
Step 2: Draw the Locus of the Generating-Circle Centre



Step 3: Divide the Fixed Circle into 12 Equal Parts



Step 4: Locate the Successive Centres



Step 5: Determine the Rolling Rotation

No-Slip Condition
When the generating circle rolls externally,

$$(R + r)\theta = r\phi$$

Therefore,

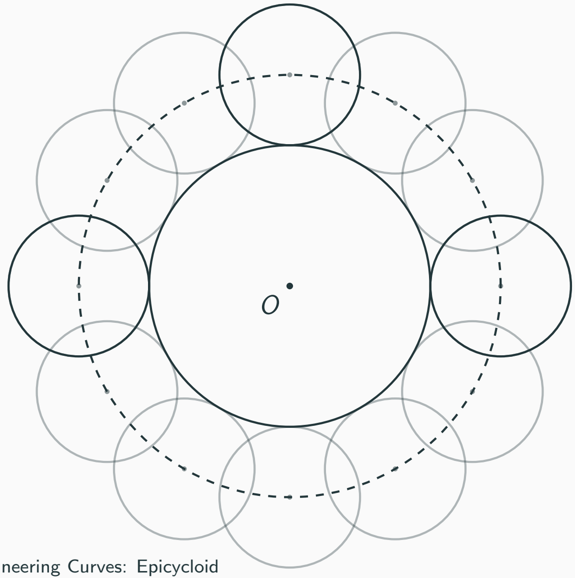
$$\boxed{\phi = \frac{R + r}{r}\theta}$$

For

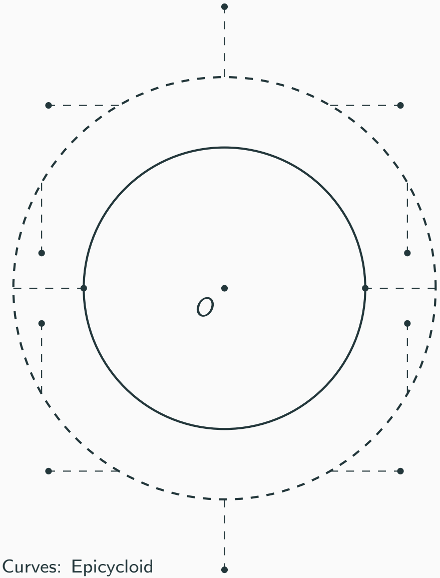
$$R = 50 \text{ mm}, \qquad r = 25 \text{ mm},$$

$$\boxed{\phi = 3\theta}$$

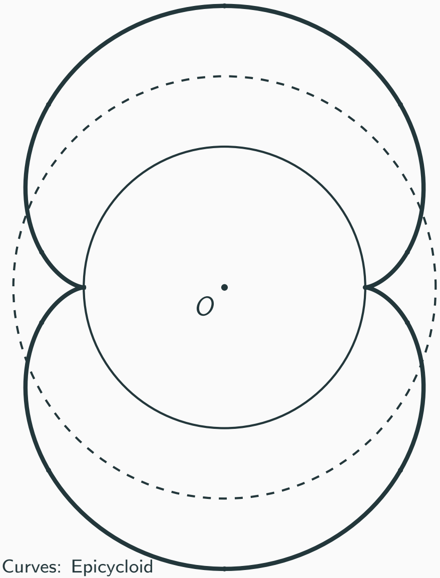
Step 6: Draw Successive Positions of the Generating Circle



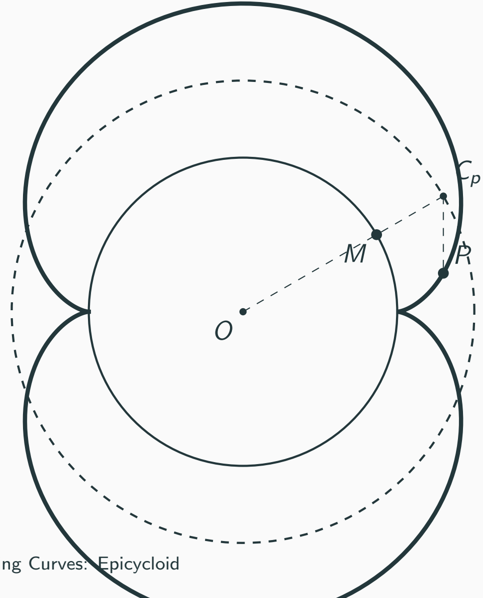
Step 7: Locate the Epicycloid Points



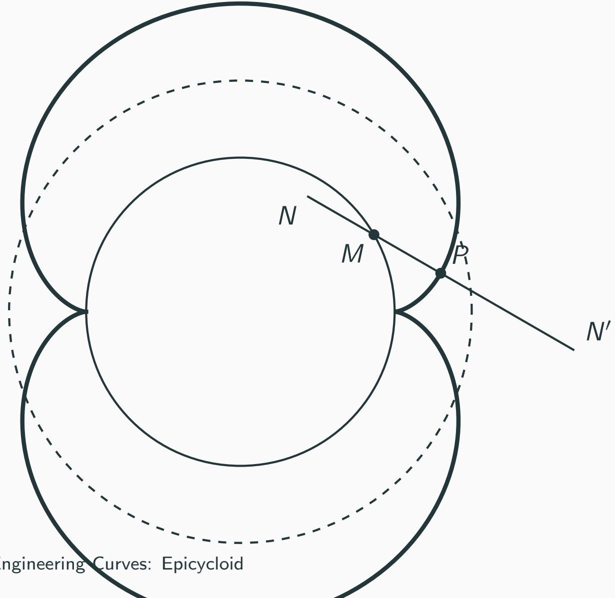
Step 8: Complete the Epicycloid



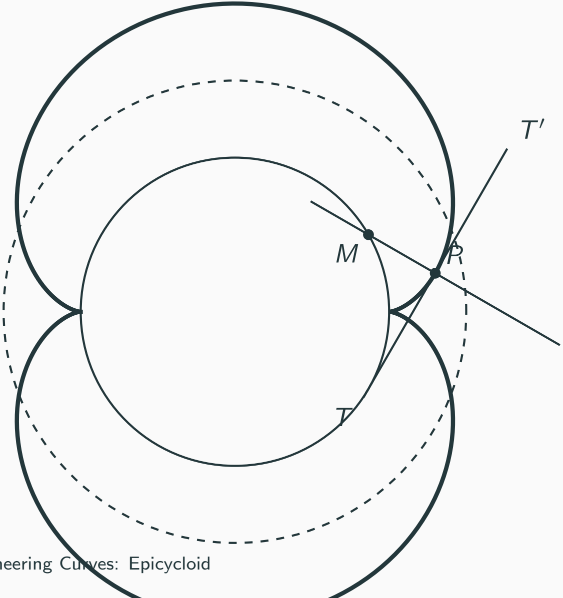
Step 9: Select Point P and Locate M



Step 10: Draw the Normal



Step 11: Draw the Tangent



Final Construction of Epicycloid